Chapter 1 Holomorpihc Functions in Several Variables

1.1 Fundamentals

1.1.1 Analysis and Topologies

Recall

- Power Series $\mathbb{C}[[z_1, \dots, z_n]]$ with elements $\sum a_{i_1,\dots,i_n} z_1^{i_1} \dots z_n^{i_n}$, $a_{i_1,\dots,i_n} \in \mathbb{C}$;
- Convergence of Power Series;
- Uniformly Convergence to function $f : \mathbb{C}^n \supseteq M \to \mathbb{C}$,

$$|\sum a_{i_1,\dots,i_n} z_1^{i_1} \dots z_n^{i_n} - f(z)| < \epsilon, \forall z \in M$$

- Holomorphic Function $\Leftrightarrow$ Analytic
  According to Cauchy's theorem in multivariables, we obtain the equivalence.

$$f \text{ is holomorphic} \Leftarrow f \text{ is analytic:} \forall z_0 \in M, \exists U(z_0), \sum a_v (z - z_0)^v \text{converges on } U(z_0) \text{ to } f(z)$$

- Absolute Space: A pointed set

$$V = \{(z_1, \dots, z_n) \in \mathbb{C}^n : z_i \geq 0\}$$

  With the modular mapping on each component,

$$\tau : \mathbb{C}^n \to V \qquad (z_1, \dots, z_n) \mapsto (|z_1|, \dots, |z_n|)$$

Covergence

Definition 1.1.1.1

- Compact Convergence: For every $K \subseteq \Omega$ compact

$$\sup_{z \in K} |f_k - f| \overset{k \to \infty}{\to} 0$$

- Locally Uniform Covergence: For every $a \in \Omega$, there exists an open neighborhood $U_a \subseteq \Omega$, such that

$$\sup_{z \in U_a} |f_k(z) - f(z)| \overset{k \to \infty}{\to} 0$$

◇ **Lemma 1.1.1.2** TFAE: For open $\Omega \subseteq \mathbb{C}^n$, $f_k : \Omega \to \mathbb{C}$

- f_k is compact covergent in Ω;
- f_k is locally covergent in Ω;

◦ **Proof**

- $\Rightarrow$ We could choose some $r > 0$, then $K = \overline{B(a, r)}$, $f_k \to f$ uniformly, thus $f_k \to f$ uniformly on $B(a, r)$.

- $\Leftarrow$ Conversely, we just select a finite covering.

□

♠ **Proposition 1.1.1.3** Let $z_0 \in \mathring{\mathbb{C}}^n$, with polycylinder $P_{z_0} = \{z \in \mathbb{C}^n || z_k| < |z_0^k| = r_k\}$, then

$$\sum_{v=0}^{\infty} a_v z^v$$

converges locally uniformly in P_{z_0}.

∘ **Proof** We just select $0 < q < 1$, then for $z \in P^* = P_{qz_0}$,

$$|z^v| = \prod |z_i|^{v_i} < q^{\sum v_i} \prod |z_0^i|^{v_i} = q^{|v|}|z_0^v|$$

Thus, we obtain the following majorance

$$|\sum a_v \cdot z^v| \leq \sum |a_v| \cdot q^{|v|} \cdot |z_0^v|$$

Now, we observe that $\{a_v z_0^v\}$ is bounded(it is the product of bounded sequence and covergent sequence to 0). So

$$\leq M \cdot \sum q^{|v|} = M \prod_1^n (\sum q^{v_i}) = \frac{M}{(1-q)^n}$$

So, the series is absolutely convergent on P_{qz_0}.
Furthermore, for arbitrary compact subset $K \subseteq P_{z_0}$, take the covering $P_{z_0} = \cup P_{q_i z_0}$. And for finite indices $q_1, \ldots, q_k$ take $q = \max\{q_i\}$. We obtain, that the series is absolutely covergent on P_{qz_0}, so it is absolutely coverent on $K \subseteq P_{qz_0}$. □

Another method is to apply T-test by Weierstrass:

1.1.2 Complex Differential and Cauchy-Riemann Equation

Definition 1.1.2.1 ...

Real Jacobian, and Standard Complex Structure

Complexification, and $(1,0),(0,1)$ **DFecomposition**

The complexifying
♠ **Proposition 1.1.2.2** $\mathbb{C}^n_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0}_{\mathbb{C}^n} \oplus V^{(0,1)}_{\mathbb{C}^n}$, where

$$V^{(1,0)}_{\mathbb{C}^n} = span_{\mathbb{C}}\{\} \qquad V^{(0,1)}_{\mathbb{C}^n} = span_{\mathbb{C}}\{\}$$

∘ **Proof** ... □

Holomorphicity

Definition 1.1.2.3 Let: $\emptyset \neq \Omega \subseteq \mathbb{C}^n$ be an open subset, then a function

$$f : \Omega \to \mathbb{C}$$

is **separately holomorphic** if for any $a \in \Omega$ and every $1 \leq j \leq n$,

$$\xi \mapsto f(a_1, \dots, a_{j-1}, \xi, a_{j+1}, \dots, a_n)$$

is holomorphic. Take the following notion

$$\mathcal{O}(\Omega) \subseteq \mathcal{O}_s(\Omega)$$

1.1.3 ∂ and $\bar{\partial}$-Operators

Definition 1.1.3.1 ...

We will discuss more in complex geometry.

1.2 Properties and Theorems

1.2.1 Cauchy's Formula

♠ **Proposition 1.2.1.1** (Cauchy) Suppose $f \in C(\bar{\Delta}(a, r))$.

If f is a function defined in a neighborhood U of a closed polydisc $\bar{\Delta}(a, r) = \prod D(a_n, r_n)$ and holomorphic in(contains) a neighborhood U and holomorphic in each variable z_i at each point U,⇔

$$f(z) = (\frac{1}{2\pi i})^n \int_{|\xi_n - a_n| = r_n} \cdots \int_{|\xi_1 - a_1| = r_1} \frac{f(\xi_1, \dots, \xi_n)}{\prod(\xi_i - z_i)} d\xi_i \dots d\xi_n, z \in \bar{\Delta}(a, r)$$

∘ **Proof** We just apply Cauchy's formula on disk componentwise. By induction

$$f(z) = \frac{1}{(2\pi i)^n} \int_{\Gamma_1 \times \cdots \times \Gamma_n} \frac{f(\xi_1, \dots, \xi_n)}{\prod(\xi_i - z_i)} d\xi_i \dots d\xi_n$$

Now, notice that the function is defined on the compact subset, thus is bounded. Fubini's theorem could be apply here. Finally,

$$= (\frac{1}{2\pi i})^n \int_{|\xi_n - a_n| = r_n} \cdots \int_{|\xi_1 - a_1| = r_1} \frac{f(\xi_1, \dots, \xi_n)}{\prod(\xi_i - z_i)} d\xi_i \dots d\xi_n, z \in \bar{\Delta}(a, r)$$

□

Shortly to say,

$$f \in A(\bar{\Delta}(a, r)) = \mathcal{O}(D) \cap C(\bar{D}) \Leftrightarrow f \in C(\bar{\Delta}(a, r)) \text{ and satisfies Cauchy integral formula.}$$

Due to Cauchy, we obtain:

★ **Corollary 1.2.1.2** Let:$\Omega \subseteq \mathbb{C}^n$ be open and $f : \Omega \to \mathbb{C}$, TFAE:

- f is complex (Frechet) differentiable at any point of Ω;
- $f \in \mathcal{O}(\Omega)$;
- (Osgood's Lemma)$f \in \mathcal{O}_s(\Omega) \cap C(\Omega)$;

∘ **Proof** By Cauchy, (I)⇔ (II)

To show (II)⇔ (III), ⇒ is easy to check

⇐ This is actually a weaker version of Osgood's lemma.

Choose an embedding disk $\bar{\Delta}(a, r) \subseteq \Omega$(regularity). The function of integral is finite on the compact subset, because of bounded.

Then, we use Cauchy's integral formula. □

This implies, that continuous separately diffentiable function is holomorphic;
And later, we will see, if $\Omega \subseteq \mathbb{C}^n, n \geq 2$,

$$\mathcal{O}_s(\Omega) = \mathcal{O}(\Omega)$$

Definition 1.2.1.3 Define: The **Cauchy integral**. Let P be a polycylinder and T the corresponding n-dimensional torus. Let f be a continuous function on T.
Then, the continuous function $ch(f) : P \to \mathbb{C}$ defined by

$$ch(f)(\xi) :=$$

Taylor's Expansion

♠ **Proposition 1.2.1.4** (Holomorphic⇔ Analytic) If f is analytic at $p \in \mathbb{C}^n$, then in a ball $B(a, r)$,

$$f(z) = \sum$$

∘ **Proof** ... □

Cauchy's Estimitation

♠ **Proposition 1.2.1.5** (Cauchy's Inequality) Suppose $f \in \mathcal{O}(\Omega)$, then we obtain the following inquality

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∘ **Proof** ... □

Identity's Lemma

▶ ***Theorem*** (Identity's Lemma)

∘ **Proof** ... □

1.2.2 Schwartz's Lemma

▶ ***Theorem*** (Schwartz) Given $\mathbb{B}^n$. And $f \in \mathcal{O}(\mathbb{B}^n, \mathbb{B}^n)$, and $f(0) = 0$, then

$$\|f(z)\| \leq \|z\|$$

∘ **Proof** Consider the Hermitian scalar product

$$h(\lambda) :=< f(\lambda u), v > \qquad u = \frac{z}{\|z\|}, v = \frac{f(z)}{\|f(z)\|}; \lambda \in \mathbb{B}^1$$

We got $|h(\lambda)| \leq |\lambda|, \lambda = \|z\| \Rightarrow \|f(z)\| \leq \|z\|$ □

1.2.3 Holomorphic, Sperately Holomorphic

Definition 1.2.3.1 Let: $f : D \times G \to \mathbb{R}$. Take the following notion

$$S_C(f) := \{(a, b) \in D \times G | f \text{ is not continuous at } (a, b)\}$$

Recall:

▶ ***Theorem*** (Baire)

∘ **Proof** ... □

★ **Corollary 1.2.3.2** (Baire)

∘ **Proof** ... □

▶ ***Theorem*** (Osgood) Suppose $\Omega \subseteq \mathbb{C}^n$ open;

- (1899) If
  - $f \in \mathcal{O}_s(\Omega)$;
  - f is locally bounded;

  Then, $f \in C(\Omega)$;
  Consequently,

$$\mathcal{O}(\Omega) = \{f \in \mathcal{O}_s(\Omega) | f \text{ is locally bounded}\}$$

- (1900) Now, suppose that $n = p+q$, with $f : \Omega \to \mathbb{C}$ such that: For every $(a, b) \in \Omega \subseteq \mathbb{C}^p \times \mathbb{C}^q$, the function

$$z \mapsto f(z, b) \qquad w \mapsto f(a, w)$$

  are holomorphic in the neighborhood of a, b respectively. Then, $S_{\mathcal{O}}(f)$ is nowhere dense in Ω;

∘ **Proof**

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□

1.2.4 Hartog's Theorem

▶ ***Theorem*** (Hartogs) Let $U \subseteq \mathbb{C}^n$ be a domain. If $n \geq 2$, $a \in U$, every holomorphic function

$$f : U - \{a\} \to \mathbb{C}$$

extends uniquely to a holomorphic function on U.

∘ **Proof** We just consider the following function: With $P(a, r) = D(a_1, r_1) \times \cdots \times D(a_n, r_n)$:

$$|z - a_1| < r_1, |\xi - z| \geq r_1 - |z - a_1| > 0 \Rightarrow F(z, w) = \frac{1}{2\pi i} \int_{|\xi - a_1| = r_1} \frac{f(\xi, w)}{(\xi - z)} d\xi \qquad (z, w) \in P(a, r)$$

And F is well-defined on $P(a, r)$; F is holomorphic:

$$\frac{\partial F}{\partial \bar{z}_i} = \frac{1}{2\pi i} \int_{|\xi - a_1| = r_1} \frac{1}{\xi - z} \cdot \frac{\partial f}{\partial \bar{z}_i} f d\xi = 0$$

Finally, $F = f$:

$$h := F - f \in \mathcal{O}(P(a, r) - \{a\}) \Rightarrow F = f \text{ on } \Omega$$

We apply identity's lemma, because $P(a, r) - \{a\}$ is connected, so there exists analytic continuation to a. □

Furthermore:

Definition 1.2.4.1 Let: $P = \{\xi \in \mathbb{C}^n | |\xi| < 1\}$ be the unit polycylinder, $q_1, \ldots, q_n$ be the real number such that $0 < q_v < 1$.
Then, define

...

Leja's Proof

Another proof is given by Leja
◇ **Lemma 1.2.4.2** ...

∘ **Proof** ... □

Koseki's Proof

Generalized Hartog's Lemma

1.2.5 Extension of Holomorphic Functions

Riemann Regions

1.2.6 Jensen's Formula

1.3 Domains of Holomorphy

1.3.1 Rheinhardt Domain

Polycylinder

Definition 1.3.1.1 Let $\tau \in V$ be an element in absolute space,$z_0 \in \mathbb{C}^n$.
then define the **polycylinder about** z_0 **with radius** τ as

$$\Delta(z_0, \tau) = \{z \in \mathbb{C}^n : |z_i - z_{0,i}| < \tau_i, i = 1, \dots, n\}$$

And, $\partial\Delta(z_0, \tau)$ is the **distinguished boundary** of $\Delta(z_0, \tau)$.

It is easy to see that the geometric shape of polycylinder is actually a **product of discs** in $\mathbb{C}^n$. Thus, the boundary is a product of circles, actually 'torus' in $\mathbb{C}^n$.

Torus Action

Definition 1.3.1.2 Let M be a smooth manifold. A **torus action** is a smooth action

$$\rho : T^n \times M \to M$$

where:

$$e \cdot p = p \qquad (st) \cdot p = s \cdot (t \cdot p)$$

♠ **Proposition 1.3.1.3** We obtain that:

- Every orbit

$$T^k \cdot p = \{t \cdot p | t \in T^k\}$$

is compact;

- The action ρ is proper;
- There exists homeomorphism by action

$$T^k \cdot p \cong T^k / T^k_p \qquad T^k_p = Stab(p) = \{t \in T^k | t \cdot p = p\}$$

∘ **Proof** ... □

For the torus action on $\mathbb{C}^n$, we have a standard action

♠ **Proposition 1.3.1.4**

- On $\mathbb{C}^n$, the torus action has a standard form

$$\rho : (z_1, \dots, z_n) \mapsto (e^{i\theta_1} z_1, \dots, e^{i\theta_n} z_n)$$

- $\mathbb{C}^n / T^n \cong [0, \infty)^n$;

∘ **Proof**

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□

Wwe will discuss more in:

- Kähler Geometry.

Definition of Rheinhardt Domain

Definition 1.3.1.5 Let $\Omega \subseteq \mathbb{C}^n$ be a domain, then D is called a **Rheinhardt Domain** iff:
It is stable under arbitrary T^n-action

$$T^n \cdot_\rho \Omega = \Omega$$

◇ **Lemma 1.3.1.6** For every Reinhardt domain Ω, we could construct an embedding

$$T^n \hookrightarrow \Omega$$

∘ **Proof** ... □

Thus, the quotient manifold structure is well-defined:

♠ **Proposition 1.3.1.7** Consider the mapping $\tau : \mathbb{C}^n \to V$. Then, we obtain the following (diffeo-?)homeomorphism

$$\Omega/T^n \cong \tau(\Omega)$$

∘ **Proof** ... □

Properness and Compeleteness

Definition 1.3.1.8 Let $G \subseteq \mathbb{C}^n$ be a Reinhardt domain:

- G is called **proper** if
  - G is connected;
  - $0 \in G$;
- G is called **complete** if

...

♠ **Proposition 1.3.1.9** ...

∘ **Proof** ... □

▶ ***Theorem*** ...

∘ **Proof** ... □

Expansion in Reinhardt Domain

1.3.2 The Continuity Theorem

Definition 1.3.2.1 Let:

- $P = \{z \in \mathbb{C}^n || z| < 1\}$;
- Real numbers $q_1, \dots, q_n; 0 < q_i < 1$;
- $D_\mu := \{z \in P || z_1| \le q_1, q_\mu \le |z_\mu| < 1\}; 2 \le \mu \le n$;
- $D := \cup_{\mu=2}^{b} D_\mu; H = P - D = \cap(P - D_\mu)$

$$= \{z \in P || z_1| > q_1 \vee |z_\mu| < q_\mu, 2 \le \mu \le n\}$$
$$= \{z \in P | q_1 < |z_1|\} \cup \{z \in P || z_\mu| < q_\mu, 2 \le \mu \le 2\}$$

Then

- (P, H) is called a **Euclidean Hartogs figure** in $\mathbb{C}^n$;
- Let (P, H) be a Euclidean Hartogs figure in $\mathbb{C}^n$, and given $g : P \to \mathbb{C}^n$ be a biholomorphic mapping, then

$$\tilde{P} = g(P) \qquad \tilde{H} = g(H)$$

such $(\tilde{G}, \tilde{H})$ is called a **general Hartogs figure**

▶ ***Theorem*** (Hartog's Extension Theorem for Hartogs Figure)

∘ **Proof** ... □

▶ ***Theorem*** (Continuity Theorem)

∘ **Proof** ... □

▶ ***Theorem*** ...

∘ **Proof** ... □

1.3.3 Pseudoconvexity

1.3.4 Holomorphic Convexity

1.4 Holomorphic Functions on Unit Ball

1.4.1 Automorphism Groups

Chapter 2 Weierstrass's Prparation Theorem

Recall: In complex analysis in variable

▶ ***Theorem***

This chapter is an introduce to the concept 'analytic'

2.1 Algebra of Power Series

Definition 2.1.0.1 Define:

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♠ **Proposition 2.1.0.2** The followings are equivalent:

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∘ **Proof** ... □

Definition 2.1.0.3 ...

Recall: **Complex Banach space**.

♠ **Proposition 2.1.0.4** ...

2.1.1 Weierstrass's Division Theorem

▶ ***Theorem*** (Weierstrass)

2.1.2 Some of Commutative Algebra Conclusions

2.2 Analytic Set

Definition 2.2.0.1 Let $B \subseteq \mathbb{C}^n$ be a region, and $M \subseteq B$ be a subsec, $z_0 \in B$ be a point.
M is called **analytic** at z_0, if there exists a neighborhood $z_0 \in U$, and $f_1, \dots, f_t \in \mathcal{O}(U)$,

$$U \cap M = \{z \in U | f_1(z) = \cdots = f_t(z) = 0\}$$

◇ **Lemma 2.2.0.2** (Closedness of Analytic Set)

∘ **Proof** ... □

Chapter 3 Analysis Tools

3.1 Lesbegue Spaces

3.2 Bergman Metric

3.3 Green's Formula

3.4 Subharmonic Functions

Now, we will introduce some of contents in the style of analysis. The knowledge like coherent sheaves usw. will be introduced in complex geometry.

The core of first part is the technique of integral kernels and function spaces.

Chapter 4 Bergman Space

Chapter 5 The Bloch Space

This part is Pluripotential Theory;

This part is harmonic method.